

Chapter – 8: Nature of Matter: Elements, Compounds, and Mixtures

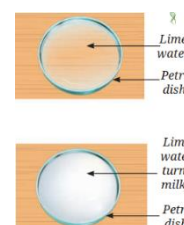
- Ever wonder – surroundings – made of?
- Look around – staircase, air, water, food, clothes, shoes, book, trees, ball, stick – all examples – **matter** – learnt – earlier grades
- Also learnt – all matter – made of – tiny particles
- Most things – surroundings – not 1 substance – rather – 2 or more – mixed together

What are Mixtures?

- Ever wonder – why *poha* delicious? – sprout salad – how to make it perfect?
- These dishes – very diff. – BUT – something common – both – mixed several ingredients
- **Observe** – mixing of substances – everyday life – sugar in water – soup, lemonade
- 2 or more substance – mixed – each constituent – retain properties – **mixture**
- Individual substances – make mixture – components
- Components – mixture – no chem. react.
- Some mixtures – components – green gram, chickpeas, onion, tomato – sprout salad – easy to spot
- Such mixtures – diff. components – visible – naked eye, magnifying glass – **non-uniform**
- **Identify** – few more examples – non-uniform mixtures
- Other hand – some mixtures – components – cannot see separately
- Such mixtures – components – even distribution – **uniform** – list few
- **A Step Further**
 - Stainless steel – mixture – iron, nickel, chromium, carbon
 - Mixed – uniformly – entire mixture – appear same – such mixtures – alloys
 - Brass – copper, zinc – bronze – copper, tin – examples – alloys
- **Interesting fact** –
 - *Mishraloha* – name – mixture – 2 or more metals – properties – diff. – constituent metals
 - Ancient Indian texts – *Charaka Samhita*, *Susruta Samhita*, *Rasaratna Samucchaya*, *Rasa Jala Nidhi*, etc. – mention alloys – medicinal purpose
 - Ex. – Bronze – *Kamsya* – alloy – Copper (*Tamra*, 4 parts) and Tin (*Vanga*, 1 part) – improve digestion – boost immunity

Is air a mixture?

- Curiosity – grade 6 – learnt – air composition – chapter – nature's treasure
- Air – mixture? – what kind?
- Learnt – air – uniform mixture – nitrogen, oxygen, argon, carbon dioxide, water vapour
- Oxygen – req. – most org. – stay alive – also helps – combustion
- Nitrogen – 78% - no part – combustion
- Air – water vapour – warm air – touch cool surface – vapour turns to liquid – tiny droplets
- Recall experiment – tested presence – carbon dioxide – exhaled air – **confirm** this
- Activity –
 - Take – glass tumbler – fill half – water
 - Add – small amt. – calcium oxide (quick lime) – slowly – obs.
 - Calcium oxide – vigorous react. – form calcium hydroxide – release heat
 - Stir continuously – make solu. – calcium hydroxide – lime water
 - Filter – obs. colour



- Leave colourless solu. – petri dish – few hrs. – keep stirring – regular intervals – obs.
- **Explain** – solu. turns milky
- Lime water – turns milky – carbon dioxide – react – calcium hydroxide – form calcium carbonate (insoluble white precipitate) and water
- Lime water – turns milky – air exposure – this act. – **demonstrate** – presence – carbon dioxide – air
 - $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
- Apart from gases – what else – present in air?
- Tiny shining particles – move around – light beam – dark room
- Activity –
 - Take – black sheet – paper – ensure – no visible dust particles
 - Place sheet – undisturbed – open window, garden – few hrs. – obs.
- Notice – tiny particles – settled – use magnifying glass – **examine** – closely
- This shows – dust particles – suspended – not integral part – pollutants
- Nature, no. – dust particles – air – vary – time, place
- **A Step Further**
 - Major pollutants – air – particulate matter – dust, soot – gases – carbon monoxide, ozone, nitrogen dioxide, sulfur dioxide
 - **Air Quality Index (AQI)** – tool – describe air quality

Types of mixtures

- Term mixture – common usage – refers – mixing – 2 or more components
- Components – themselves mixture – *poha*, sprout salad – OR – pure substances – sugar, common salt in water
- Science – all components – mixtures – pure substances
- Mixtures – several types – depend – phy. state – components
- Some mixtures – examples – shown

S. No.	Mixture Type	Examples
1.	Gas and gas	Air
2.	Gas and liquid	Aerated water (soda water), Oxygen dissolved in water
3.	Solid and gas	Carbon particles in air
4.	Liquid in liquid	Acetic acid in water (vinegar), Oil and water
5.	Solid and liquid	Sand and water, Seawater
6.	Solid and solid	Baking powder (baking soda and tartaric acid), Alloys

- Learnt – earlier grades – separation – mixtures – separate components
- Examples – everyday life – separation done – obtain components – interest – other components – discarded
- Science – purpose – separation – obtain pure substances

What are Pure Substances?

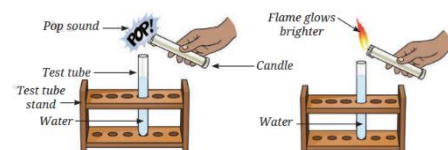
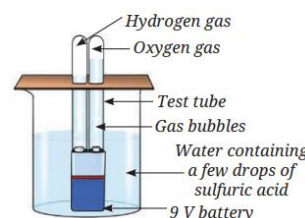
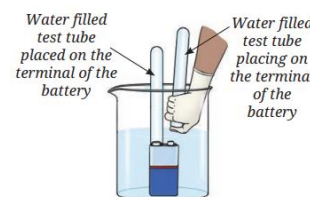
- Ever notice – pure – packs – consumables – milk, ghee, spices
- Pure – diff. meaning – common usage, science
- Common usage – **pure** – unadulterated products
- **Adulteration** – illegal process – add substances – cheaper, poor quality – done – increase quantity, reduce cost – BUT – adulteration – reduce quality – also make – product hazardous

- Science – pure substance – no other substance present
- Scientist – pure looking substances – **impure** – made of – 2 or more components
- Pure **substance** – matter – cannot separate – other kinds – matter – phy. processes
- Scientist say – something pure – substance – same kind – particles

What Are the Types of Pure Substances?

- Recall – diff. states – water – learnt – curiosity – grade 6
- Water – cooled, heated – obs. – cooling – convert to ice – boiling – convert to vapour
- Heat ice, cool vapour – get water back
- During processes – particles of water – remain same
- Activity –

- Collect – 2 small test tubes, beaker, 9 V battery
- Fill beaker – 2/3rd – water – add few drops – dil. H_2SO_4
- Fill test tube – solution from beaker
- Place battery – beaker
- Without spilling – place test tubes – each terminal – battery
- Wait – few minutes – obs. – gas bubbles – both terminals
- Continue – 10-15 min.
- Obs. vol. – gas – both test tubes – diff. levels
- Remove test tubes – carefully
- Test gases – bring burning candle – close to mouth – test tube



- Test gases – bring burning candle – test tube mouth
- One test tube – pop sound – indicate hydrogen gas
- Other test tube – flame – brighter – indicate oxygen gas
- This act. – **infer** – water – 2 components – hydrogen, oxygen
 - Water → Hydrogen + Oxygen
- **Interesting fact** –
 - Elec. current – pass through water – breaks – hydrogen, oxygen
 - This change – phy. / chem. – recall – curiosity – grade 7 – chapter – changes

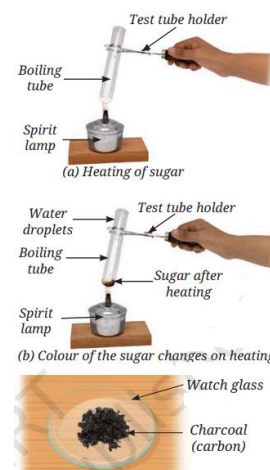
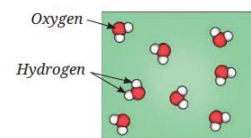
Elements

- 2 substances – Hydrogen, Oxygen – above act. – pure substances – termed – **elements**
- Each element – made of – identical particles – atoms – diff. – particles – other elements
- Elements – substances – cannot break further – simpler substances – building blocks – all matter
- Other examples – elements – gold, silver, sulfur, carbon
- **A Step Further**
 - Atoms – most elements – no independent existence
 - 2 or more atom – combine – stable particle – molecule
 - Example – 2 atoms – hydrogen – combine – 1 molecule
 - 2 atoms – oxygen – combine – 1 molecule
- Elements – classified – **metals, non-metals**
- Already learnt – gold, silver, magnesium, iron, aluminium – metals – carbon, sulfur, hydrogen, oxygen – non-metals
- Interesting – some elements – silicon, boron – intermediate prop. – **metalloids** – learn – higher classes
- **A Step Further**
 - No. of elements – known present – 118 – most – solid state

- 11 elements – gaseous state – room temp. – all – non-metals
- 2 elements – liquid – room temp. – mercury – metal – bromine – non-metal
- Gallium, cesium – solids – become liquid – temp. – 30 °C (303 K)
- **A Step Further**
 - 45+ elements – aluminium, copper, silver, silicon, cobalt, lithium, gold, silver, etc. – used – manufacture mobile

Compounds

- Cannot separate – hydrogen, oxygen – water – phy. means
- Water – particles – hydrogen, oxygen – tightly attached – each other – gen. impossible – separate them – phy. methods – reason – water – compound
- **Compounds** – formed – diff. elements combine – fixed ratio – form something new
- Prop. – compounds – diff. – constituent elements
- Constituent elements – cannot separate – phy. methods
- Above act. – molecules – water – 2 different elements – hydrogen, oxygen – fixed ratio – chem. combination
- Ratio – no. of atoms – hydrogen, oxygen – found – 2 : 1
- Sodium – soft metal – chlorine – toxic gas – combine – harmless, tasty substance – essential – sodium chloride – ratio – 1 : 1
- Learnt – dissolved salt – separated – phy. method – evaporation
- Activity –
 - Put – teaspoon sugar – boiling tube
 - Heat – gently – obs.
- Heat sugar – turns brown – chars – turns blackish
- Find – small drops – water – boiling tube – open end
- Water – origin? – dry sugar? – OR – condensation?
- Tube – heated – water – origin – dry sugar – NOT air
- **Predict** – what's left?
- Charcoal (carbon) – left behind – boiling tube – scoop out – watch glass – explore – if burns like coal
- Sugar decompose – heating – result – carbon – water – hydrogen, oxygen
- Sugar – not element – BUT – chem. compound – elements – carbon, hydrogen, oxygen
- Activity –
 - Take – 5.6 grams – iron fillings – 3.2 grams – sulfur powder – watch glass – obs.
 - Mix thoroughly – watch glass – label mixture – Sample A – obs.
 - This mixture – uniform, non-uniform? – obs. both – iron, sulfur – separately
 - Take half – sample A – china dish – heat gently – continuous stirring – until black mass
 - Let cool – content – china dish
 - Place black mass – mortar – grind – pestle
 - Put it – another watch glass – label – Sample B
 - Now – 2 samples – A, B – **Compare** – step by step – **record** obs.
 - Step 1 – appearance
 - Compare appearance – A, B – colour, texture
 - Step 2 – magnet test
 - Take magnet – move over – A, B – obs.
 - Step 3 – gas test



- Take – small amt. – sample A – test tube – add few drops – dil. HCl – obs.
- Smell – evolved gas – wafting towards nose
- Test – evolved gas – lit candle – test tube mouth
- Repeat – sample B
- Discussion points
 - Samples – A, B – look same?
 - Which sample – magnetic prop.?
 - Separate components – A, B?
 - Add dil. HCl – gas evolve – both samples?
 - Both cases – gas smell – same, different?
 - **Categorize** – substances – this act. – mixtures, compounds, elements
- Sample A –
 - Mixture – 2 elements – iron, sulfur
 - Components – retain prop. – black, yellow particles – visible
 - Bring magnet – sample A – iron fillings – attracted – iron, sulfur – separated
 - Obs. – sample A – mixture – iron – react – dil. HCl – form iron chloride, hydrogen gas
 - Gas – colourless, no less – burns – pop sound
 - Reaction –
 - $\text{Iron} + \text{dil. HCl} \rightarrow \text{iron chloride} + \text{hydrogen gas}$
 - Sulfur – left – yellow solid – bottom – test tube – shows – sulfur – no reaction – HCl
- Sample B –
 - Black mass – sample B – iron sulphide
 - Obs. – texture, colour – same throughout – formed on heating – 2 elements – iron, sulfur
 - Magnet – no effect – new substance – complete diff. prop. – iron, sulfur – cannot separate
 - Thus – compound formed
 - Also – iron sulfide – react – dil. HCl – form – iron chloride, hydrogen sulfide gas
 - Gas – colourless – rotten-egg like odour
 - Reaction –
 - $\text{Iron sulfide} + \text{dil. HCl} \rightarrow \text{iron chloride} + \text{hydrogen sulfide}$

How Do We Use Elements, Compounds, and Mixtures?

- Elements, compounds, mixtures – all around
- Air – mixture – gases – oxygen, nitrogen, carbon dioxide
- Water – essential – life – compound – elements – hydrogen, oxygen
- Elements – iron, aluminium – construct bridges, buildings, vehicles
- Understand concepts – NOT only recognizing – BUT also innovating
- Chemists – study – elements combine – create compounds – enable – invent – medicines, vaccines – fight diseases
- This knowledge – helps – creation – fertilisers – enhance crop production – feed – increasing population
- Engineers, material scientists – rely – understanding – compounds, mixtures – design materials – unique prop.
- Example – alloys – stainless steel – stronger, more durable – pure iron
- Wool, steel, concrete – building materials – all mixtures
- Learnt – various metals – obtained from minerals
- **A Step Further**
 - Example – wonder material – graphene aerogel

- Made of – carbon – lightest material – so light – grass can hold
- Highly porous – high absorbing capacity – this reason – used – environment cleaner
- Example – oil spill – seas, land
- Useful – fabricating energy-saving devices, special coatings – buildings

What Are Minerals?

- Most rocks – mixture – **minerals** – visible – naked eye, magnifying glass, microscope
- Some minerals – native minerals – pure elements – not compounds
 - Metals – gold, silver, copper, etc.
 - Non-metals – sulfur, carbon, etc.
- Most minerals – compounds – 2 or more elements
- Common examples – quartz, calcite, mica, pyroxene, olivine
- Many things – everyday life – made of – minerals, elements
- Example – cement – made of – calcite, quartz, alumina, iron oxide – minerals – OR – obtained from minerals
- Talcum powder – made of – talc – mineral
- **Interesting fact –**
 - Dhokra art – old craft – Bihar, Odisha – use diff. metals – create beautiful figurines
 - Process start – shape design – beeswax – wax model – covered in clay – make mould
 - Clay hardens – wax melted out – leave hollow space
 - This space – filled with – molten brass, bronze – Dhokra art – strong, shiny golden colour
 - Figurines – depict – animals, people, nature – reflect tribal creativity, tradition
- Elements, compounds – building blocks – matter – everything – mass, space
- Make materials – see, use everyday
- Everything around – not matter
- Light, heat, elec., thoughts, emotions – imp. parts – our world – BUT – not matter
- Understand – what matter is? – what matter is not? – understand world – better way

